

Econ 139: Human Capital

David Arnold



Human Capital

- The capital of a firm includes all of its resources at its disposal to produce output. This could include financial capital or physical capital like machines
- Human capital is the analogous concept for workers: the sum total of a worker's skills is their human capital
- In this chapter, we will discuss how human capital is formed, mainly studying returns to various forms of education

Human Capital and Growth

- A primary goal of macroeconomics is to understand causes of growth
- Human capital is key in these frameworks: innovation drives growth. Increases in human capital drive innovation

Human Capital and Growth

- Places with higher levels of schooling have higher levels of output per capita
- Size of dots represent population, colors represent region

Men's Schooling and GDP Per Capita

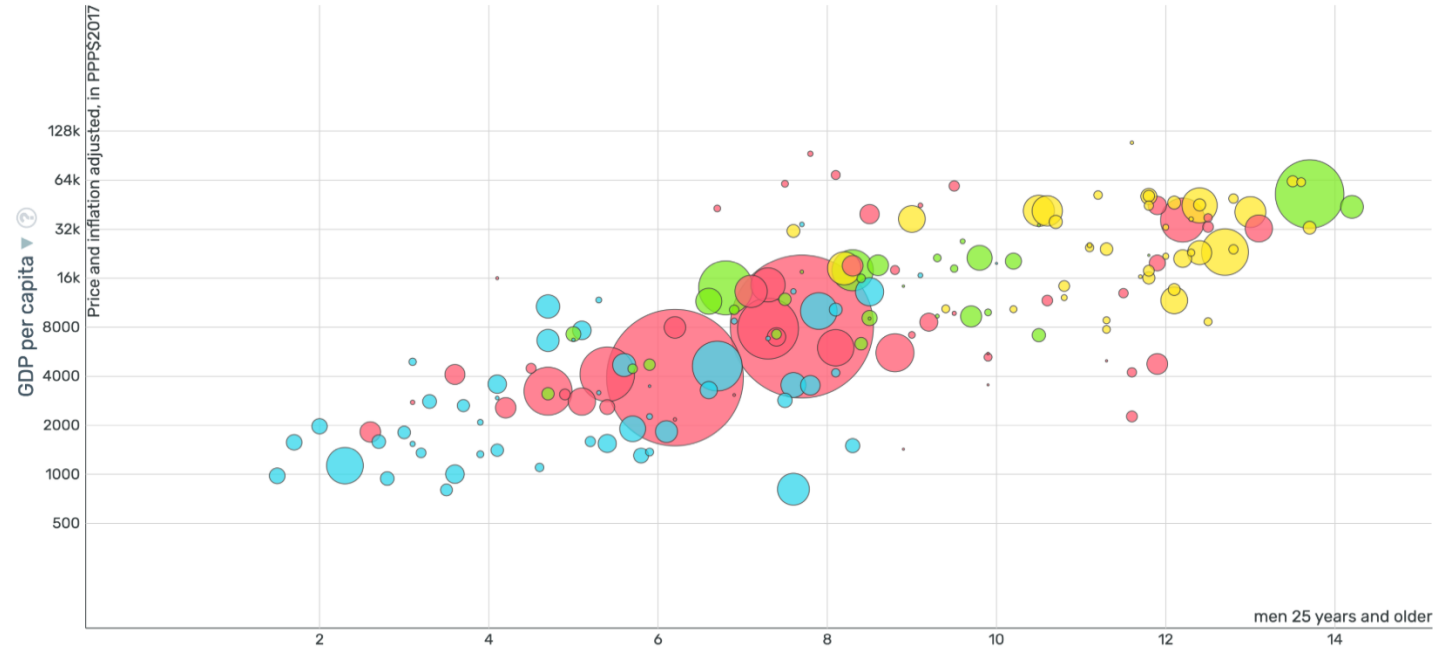
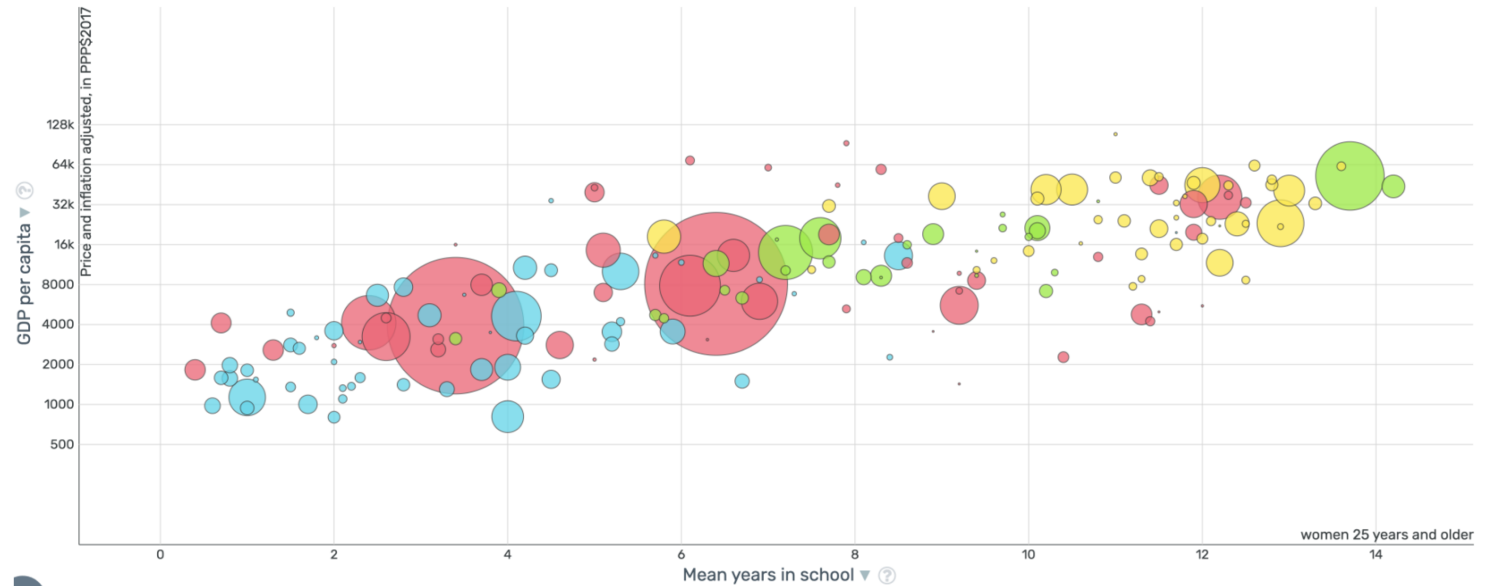


Figure 6.1: GDP Per Capita and Mean Schooling_(Men).

Human Capital and Growth

- Places with higher levels of schooling have higher levels of output per capita
- Size of dots represent population, colors represent region

Women's Schooling and GDP Per Capita



[Figure 6.2: GDP Per Capita and Mean Schooling \(Women\)](#)

Human Capital and Growth

- Data like this has played a huge role in policy decision
 - If you want to have a more productive economy, it pays to invest in education
 - There are also alternative stories/other considerations: if you want a more productive economy, have institutions that make it possible for people to gain from their educational choices
- We can also ask within a country, how much do individuals benefit in terms of labor-market returns for additional years of schooling

Human Capital Theory (Becker 1964)

- Gary Becker is a Nobel-prize winning economist famous for his many creative contributions to the field of Economics
- One of his more notable works is called Human Capital, and it includes a framework for analyzing education decisions by individuals in the same way we analyze investment decisions in firms
 - Instead of firms deciding whether to not invest in physical capital, we are considering what drives individuals whether or not to invest in human capital through further education

Framework

- Let Y_t^1 be the amount you earn in time period t if you go to college
- Let Y_t^0 be the amount you earn in time period t if you don't go to college
- If $Y_t^1 > Y_t^0$, then going to college raises the amount you earn in time period t

Framework

- Now imagine you are an 18 year old deciding whether to go to College or not
- You will consider the costs – you will need to pay for tuition now, you will have lower earnings for several years
- You will also consider the benefits – in the future you expect to earn more
- So there is an initial period of lower earnings, followed by a period of higher earnings? How do we decide which stream of income is “better”?

Net Present Value

- This setup is actually quite common in investment decisions
- Imagine a firm is deciding whether to buy a machine or not.
- If the firm buys the machine, they will have lower income this period, but higher in the future. For example, let's assume the firm operates for only three years and will have the following income stream if they buy the machine

$(0, 100, 100)$

Net Present Value

- If the firm decides not to buy the machine, then they will earn 20 today, 20 in the next year and 20 in the final year

$(20, 20, 20)$

- Is $(0, 100, 100)$ better than $(20, 20, 20)$? The key for us in deciding will be to compute the **net present value** of both income streams

Net Present Value

- To understand net present value, let's consider an isolated part of this problem. What is the value of 100 dollars a year from now?
- One way to answer this is: if I had X number of dollars today, I could put it in the bank and get some interest on this money.
- If the total amount of money equals 100, then I am indifferent between X today and 100 dollars tomorrow
 - X today immediately implies I will have 100 tomorrow

Net Present Value

- If the interest rate is 10 percent, then how much do I need to have today so that I can ensure I receive 100 in the following year?

$$X * (1.10) = 100$$

$$X = \frac{100}{1.10} = 90.91$$

- Said another way: if I have 90.91, I can put that in the bank. I will receive 9.09 in interest. So if I add up the original plus interest I will have 100 tomorrow

Net Present Value of 100 in two years

- If the interest rate is 10 percent, then how much do I need to have today so that I can ensure I receive 100 in **two years**?
- Again, how much do I need today so that in two years, I would have 100 dollars
- If I had 82.64 today, I could invest at 10 percent and get $82.64 + 8.27 = 90.91$. We already know the rest. If I invest 90.91 at a rate of 10 percent interest I get 100 in another year

Net Present Value of Income Stream

- The net present value of the entire income stream of (0,100,100) is equal to:

$$NPV = 0 + \frac{100}{1.1} + \frac{100}{(1.1)^2} = 173.55$$

- The general formula for a NPV of a generic income stream of (Y_0, Y_1, ...Y_N) is:

$$NPV = \sum_0^N \frac{Y_t}{(1+r)^t}$$

So which income stream is better?

Buy Machine

$(0, 100, 100)$

Don't Buy
Machine

$(20, 20, 20)$

So which income stream is better?

Buy Machine

(0,100,100)



$$0 + \frac{100}{1.1} + \frac{100}{(1.1)^2} = 173.55$$

NPV

Don't Buy
Machine

(20,20,20)



$$20 + \frac{20}{1.1} + \frac{20}{(1.1)^2} = 54.71$$

Human Capital Theory Conceptualizes Decision for Education in Same Way

College

(0,100,100)



$$0 + \frac{100}{1.1} + \frac{100}{(1.1)^2} = 173.55$$

NPV

No
College

(20,20,20)



$$20 + \frac{20}{1.1} + \frac{20}{(1.1)^2} = 54.71$$

Becker's Formulation

- Becker's theory implies education choices depend on two parameters
 - Interest rates (also called discount rate)
 - Returns to Education

$$NPV = \sum_{t=0}^T \frac{Y_t^1 - Y_t^0}{(1 + r)^t}$$

- This equation writes the NPV as the difference between receiving education and not receiving education. If it is positive, then the individual will pursue further education

Our Goal: Returns to Education

- Our analysis will focus much more on the returns to education ($Y_t^1 - Y_t^0$)
- To simplify things, let's ignore time periods and denote the average return to education across your career as ($Y^1 - Y^0$)
- While our illustrative example has considered this as college vs. no college, this is more general. You could think about this as deciding to pursue another year of education vs. not
- An enormous literature in labor economics has sought to estimate this value. We will discuss this literature next

Mincer's Equation

- One of the most famous uses of Becker's theory is the derivation of the Mincer Equation, which relates an individual's earnings to a few pretty simple variables
- These variables include the level of education (in years), the experience of the worker, and the experience of the worker squared

Mincer's Equation

$$\ln(w) = \alpha + \beta_1 S + \beta_2 E + \beta_3 E^2 + \varepsilon$$

Where:

- w = earnings (wage or income)
- S = years of schooling/education
- E = years of work experience (often approximated as age - schooling - 6)
- α = intercept term
- β_1 = return to education (approximately the percent increase in earnings per additional year of schooling)
- β_2, β_3 = parameters capturing the relationship between experience and earnings
- ε = error term

Mincer's Equation

- The goal of the Mincer equation is to explain variation in wages across individuals using a very simple list of variables
- To understand how it is used and what it implies, we are first going to consider the terms of experience and experience squared
- To see why we have experience squared in this equation, we will first show what omitting experience squared would imply, and why this is at odds with the data

Mincer's Equation

- The Mincer Equation without experience squared would be:

$$\ln(w) = \alpha + \beta_1 S + \beta_2 E + \varepsilon$$

- The form of this implies that individuals gain the same amount for every 1-year increment in experience?
- E.g. if $\beta_2 = 0.1$, then this equation predicts wages increase by 0.1 log points (roughly ten percent), every single year. This is true going from 0 years to 1 year of experience and 19 to 20 years of experience
- We can look in the data and see if this is true

The American Community Survey

- The American Community Survey (ACS) is another household survey run by the Census
- It collects many of the same variables as the CPS
 - Income, demographics, education, etc.
- However, the yearly sample size is quite a bit larger. The CPS surveys about 60,000 unique households a year (though they interview each household 8 times). The ACS sample size is about 3 million households

Earnings and Experience in the 2023 ACS

- We are going to use the 2023 ACS to understand the relationship between earnings and experience
- Experience is generally measured as

$$\textit{Experience} = \textit{Age} - \textit{Years of Schooling} - 6$$

- Assumes individuals start school at age 6, then enter workforce directly after completing their education



Figure 6.3: Experience and Annual Wage Income

Annual Wage
Income and
Experience in
the 2023
ACS

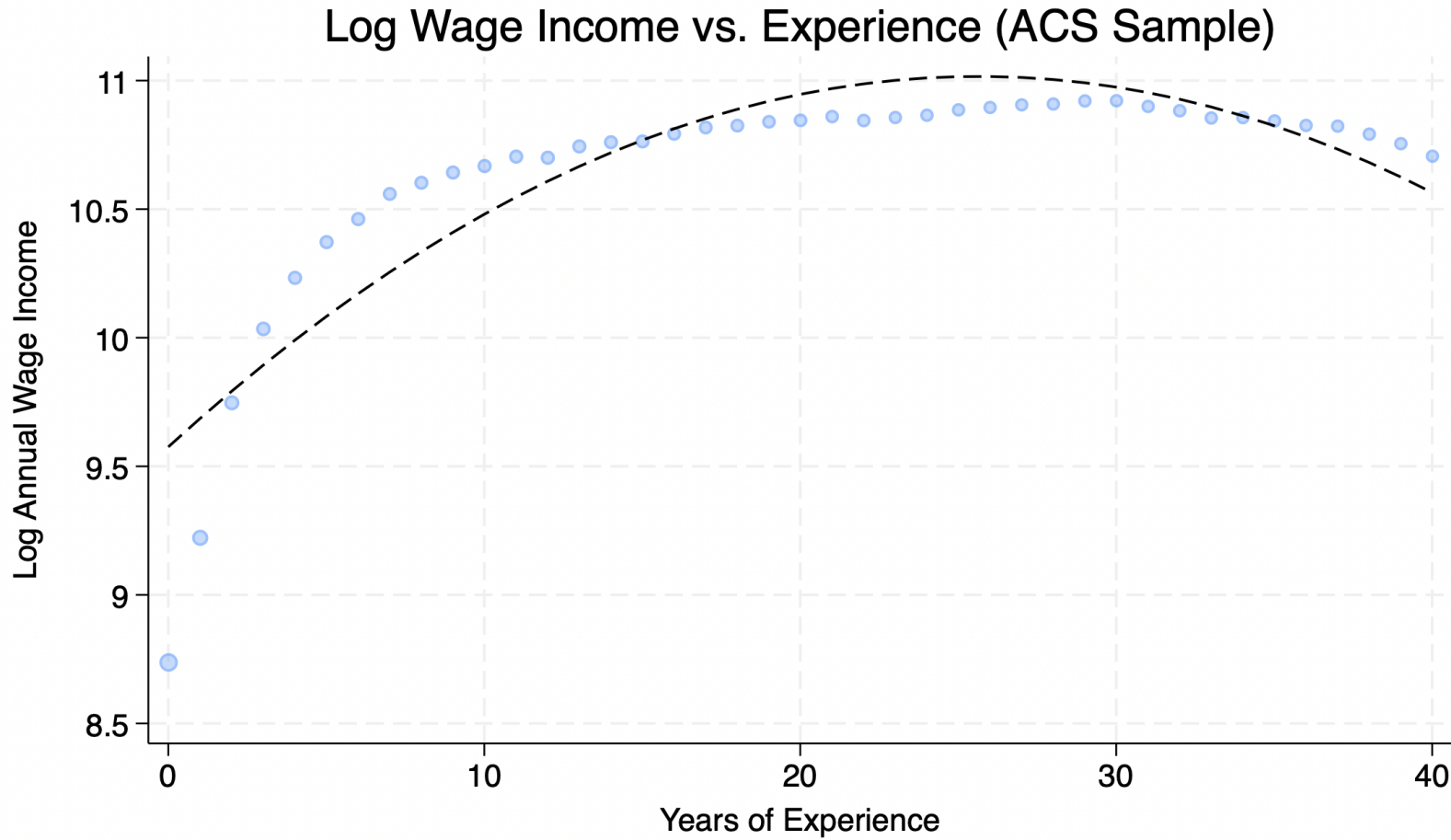


Log Annual
Wage
Income and
Experience in
the 2023
ACS

Experience

- So initially there are very high returns to experience, but these returns fade over time
- This cannot be captured by a model with only a linear term in experience
- However, it can be modelled by a quadratic equation

$$\ln(w) = 9.6 + 0.112 \cdot E - 0.002 \cdot E^2$$



Log Annual
Wage
Income and
Experience in
the 2023
ACS

Schooling

- Now, that we understand why experience squared is in the equation let's return to our main goal: what is the return to an additional year of schooling
- To explore this, we are first going to plot average log earnings by years of schooling, which we can observe in the ACS

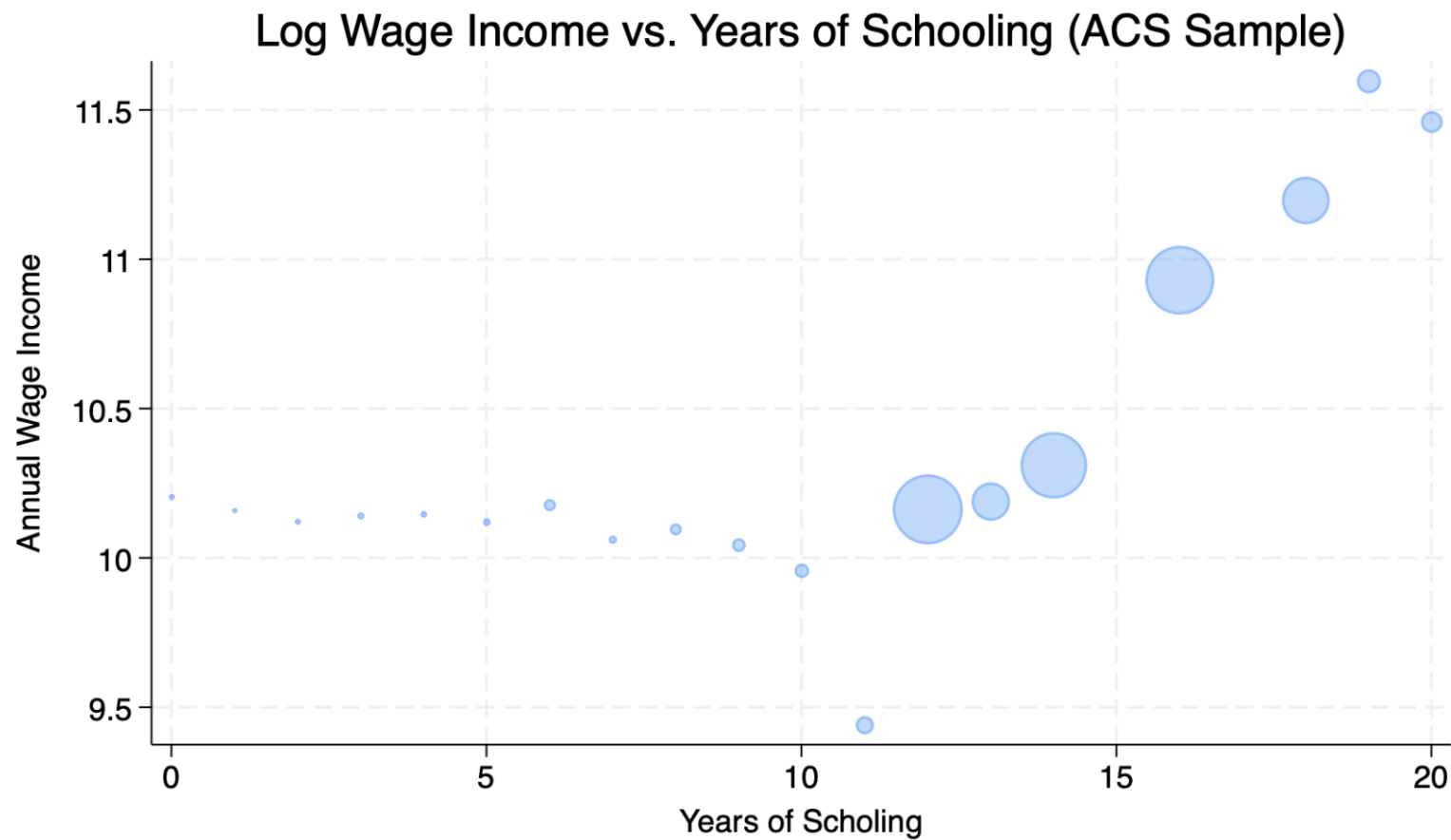
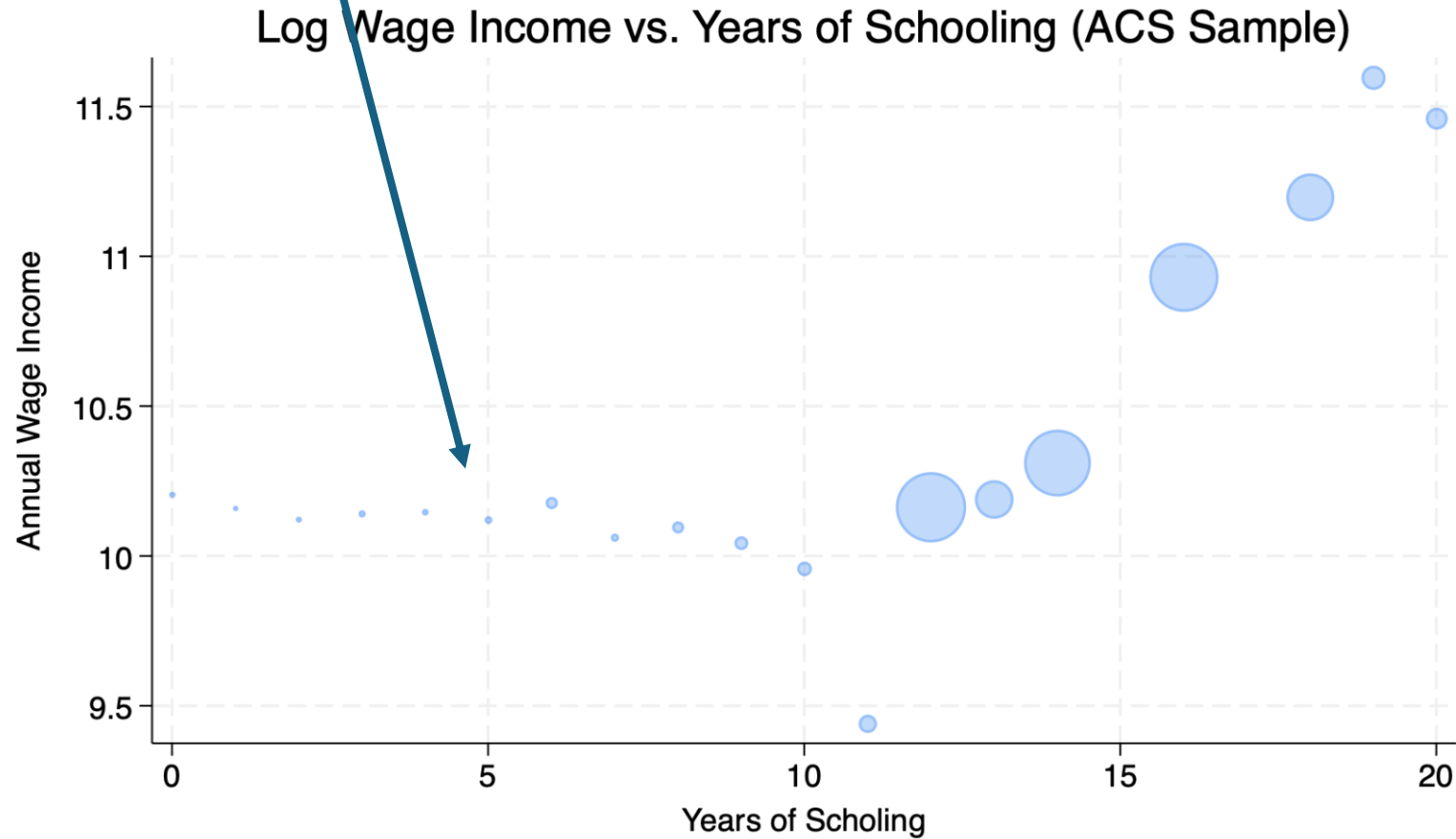


Figure 6.6: Years of Schooling and Log Annual Wage Income

Log Annual
Wage
Income and
Schooling

Size of dots reflect number of observations with that level of schooling



Log Annual
Wage
Income and
Schooling

Figure 6.6: Years of Schooling and Log Annual Wage Income

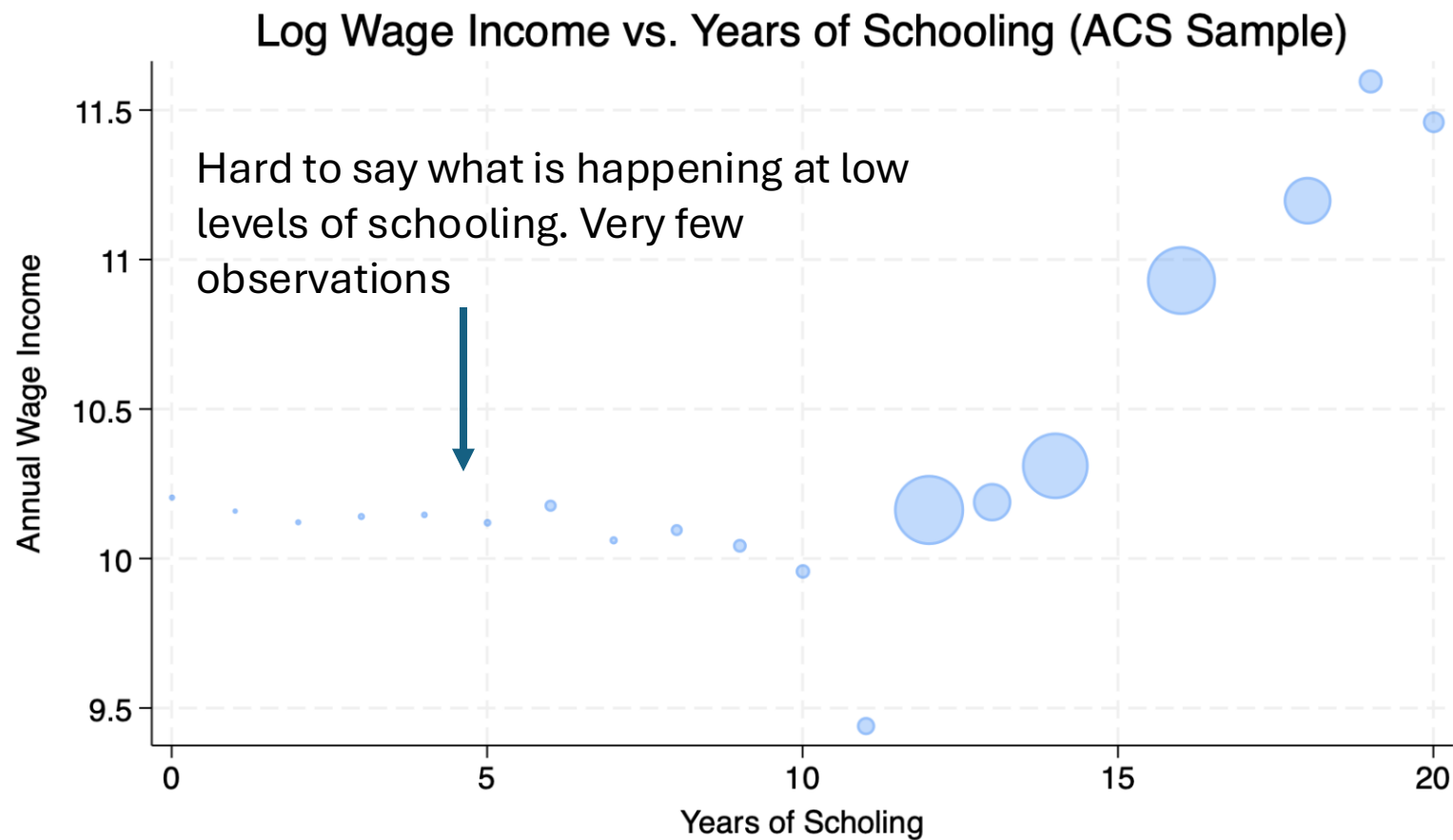


Figure 6.6: Years of Schooling and Log Annual Wage Income

Log Annual
Wage
Income and
Schooling

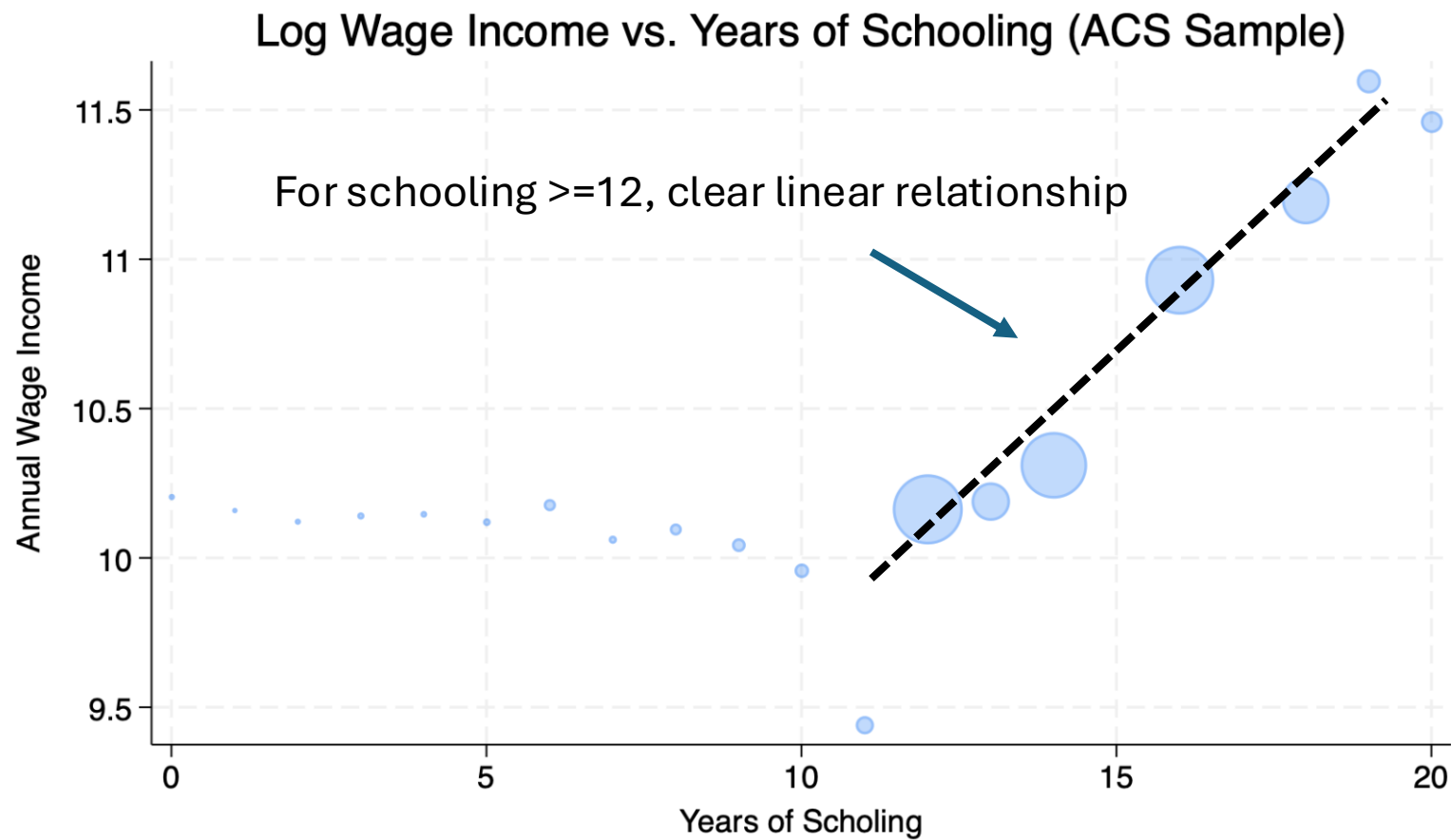


Figure 6.6: Years of Schooling and Log Annual Wage Income

Log Annual
Wage
Income and
Schooling

Putting it all together

- To estimate the Mincer Equation, generally one would use Ordinary Least Squares to choose the parameters that “best-fit” the data
 - This means choosing the betas that minimize the sum of squared residuals (because Econ 120A is not a prerequisite for this class, we will not be digging into the details)
 - Instead we will show what an estimated Mincer Equation would look like and discuss how to interpret the parameters

Putting it all together

A typical estimated Mincer equation might look like:

$$\ln(w) = 7.069 + 0.173 \cdot S + 0.100 \cdot E - 0.002 \cdot E^2 + \varepsilon$$

This tells us for a given level of schooling and experience, what we expect an individual's earnings will be

When making predictions you assume the error term is zero (this will be true on average due to how we estimated the parameters of the model, but you don't need to worry about the nuances of this for this course)

Predicting Wages

A typical estimated Mincer equation might look like:

$$\ln(w) = 7.069 + 0.173 \cdot S + 0.100 \cdot E - 0.002 \cdot E^2 + \varepsilon$$

What does this equation suggest the wage will be for someone with 16 years of schooling and 10 years of experience

Predicting Wages

A typical estimated Mincer equation might look like:

$$\ln(w) = 7.069 + 0.173 \cdot S + 0.100 \cdot E - 0.002 \cdot E^2 + \varepsilon$$

What does this equation suggest the wage will be for someone with 16 years of schooling and 10 years of experience

$$\ln(w) = 7.069 + 0.173 \times 16 + 0.100 \times 10 - 0.002 \times 10^2$$

$$\ln(w) = 7.069 + 2.768 + 1.000 - 0.200 = 10.637$$

Predicting Wages

Now imagine we consider another person who has the same experience, but has 17 years of schooling (i.e. they went to graduate school for one year). Their predicted log wage is equal to:

$$\ln(w) = 7.069 + 0.173 \times 17 + 0.100 \times 10 - 0.002 \times 10^2$$

$$\ln(w) = 7.069 + 2.941 + 1.000 - 0.200 = 10.810$$

So the difference in log wages between these two individuals is $10.810 - 10.637 = 0.173$. So

What does this mean

- There is a simple formula that allows us to translate log difference to percent differences

$$\% \Delta w \approx 100 \times \exp(\beta_1) - 1$$

- For us this implies

$$\% \Delta w \approx 100 \times \exp(0.173) - 1 \approx 18.9\%$$

So 1 year of schooling is predicted to increase your wages by 18.9%!

Correlation vs. Causation

- So far we have documented a strong **correlation** between schooling and earnings
- But who is to say any of this is causal
 - Maybe more motivated individuals pursue more education. It is the motivation that is causing higher earnings, not the education itself
- In the next set of slides we will discuss methods that have been proposed to tease out the **causal impact** of education on earnings